Physics 23: Electromagnetism and Waves

Practice Midterm: Complete Answer Key

Spring 2026 — Purcell Chapters 1 & 2

FOR INSTRUCTOR / LEARNING ASSISTANT USE ONLY

Problem 1

[18 points]

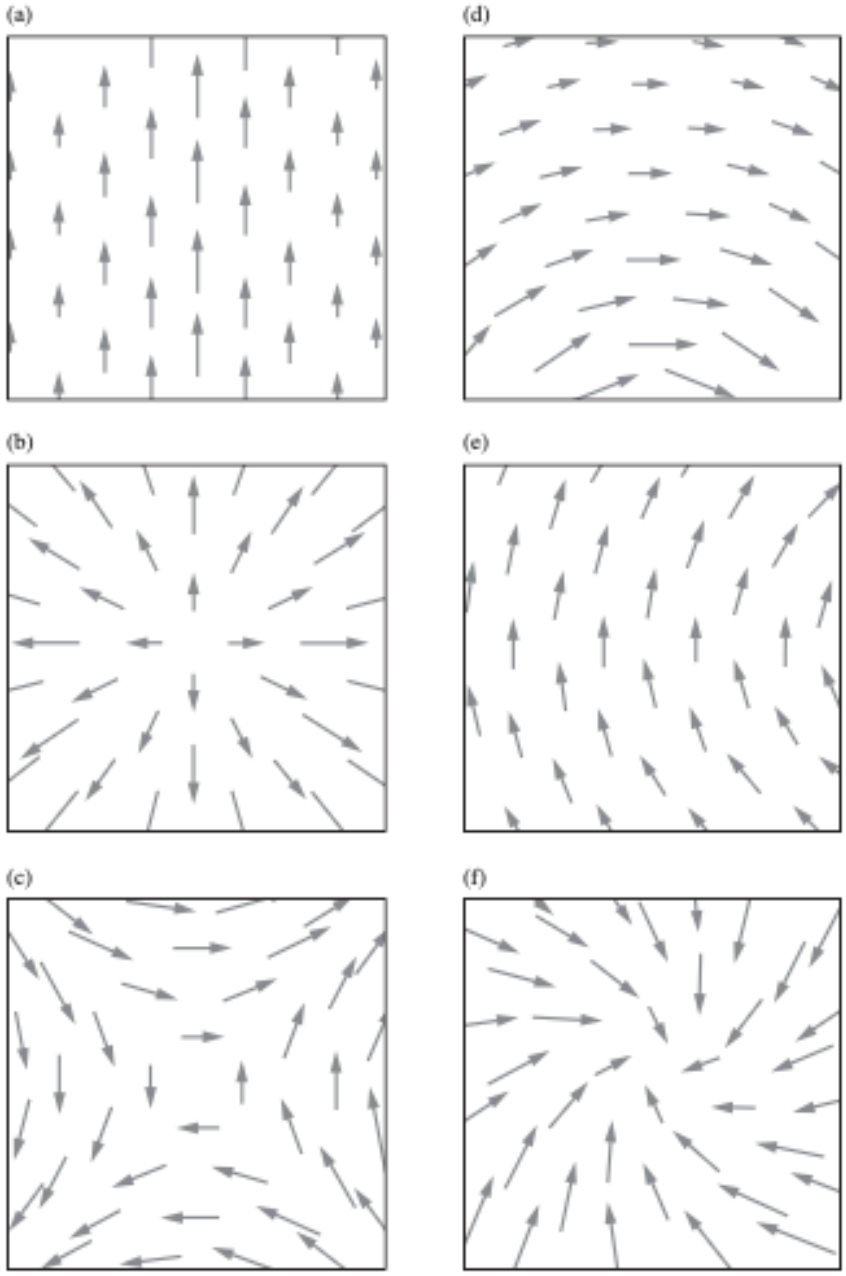
Reading Vector-Field Diagrams

The six panels (Purcell Figure 2.39) show two-dimensional vector fields. For each panel, students must determine:

(i) Is $\nabla \cdot \vec{E} = 0$ in the region shown?

(ii) Is $\nabla \times \vec{E} = \vec{0}$ in the region shown?

(iii) Could this field be a static electric field in a charge-free region?



Key physical rules to apply:

- $\nabla \cdot \vec{E} = 0$ iff field lines neither start nor end in the region (no net flux out of any tiny volume). Visually: field line density does *not* change — lines don't converge, diverge, or terminate.
- $\nabla \times \vec{E} = \vec{0}$ iff there is no net circulation around any tiny closed loop. Visually: the field does *not* rotate or curl around any point.
- A valid electrostatic field in a charge-free region requires **both**: $\nabla \cdot \vec{E} = 0$ **and** $\nabla \times \vec{E} =$

$\vec{0}$.

Solution

Panel	(i) $\nabla\cdot\vec{E}=0$?	(ii) $\nabla\times\vec{E}=\vec{0}$?	(iii) Valid?
(a)	**Yes.** Uniform field — line density constant everywhere, no sources or sinks.	**Yes.** No circulation; all arrows parallel, no looping.	**Yes.** Both conditions met — uniform field in a vacuum is valid.
(b)	**No.** Field lines spread radially outward — net flux out of any small volume centered near the source is nonzero ($\nabla\cdot\vec{E}\neq 0$ at the source).	**Yes.** Purely radial field has no circulation.	**No.** Nonzero divergence implies $\rho\neq 0$; not charge-free.
(c)	**Yes.** Magnitude is constant along circular loops; no net flux escapes any closed surface.	**No.** Field curls around a center — a small loop aligned with the field has nonzero circulation, so $\nabla\times\vec{E}\neq\vec{0}$.	**No.** $\nabla\times\vec{E}\neq\vec{0}$ is forbidden for a static electric field.
(d)	**No.** Field lines get denser in one direction — the magnitude increases along the flow, so $\partial E_x/\partial x\neq 0$, giving a nonzero divergence.	**Yes.** Arrows point in a single direction with no curl.	**No.** Nonzero divergence implies charge density present.
(e)	**Yes.** Field lines neither start nor end; the pattern (e.g., saddle or hyperbolic) can have $\nabla\cdot\vec{E}=0$ if components cancel: $\partial E_x/\partial x+\partial E_y/\partial y=0$.	**Yes.** No rotational component; the field is irrotational.	**Yes.** Both conditions satisfied — this can represent the field near a saddle point of an electrostatic potential.
(f)	**No.** Field spirals inward (or outward) — field lines converge toward (or diverge from) the center, giving net nonzero flux through small surfaces.	**No.** Spiral pattern has rotational component; net circulation around a small loop is nonzero.	**No.** Fails both conditions.

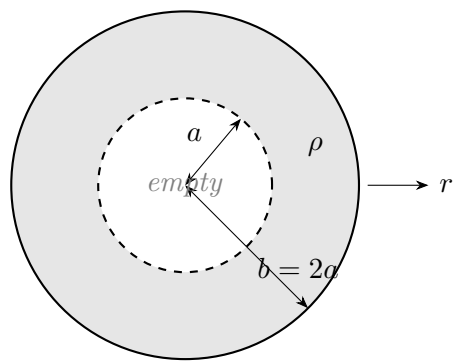
Grading note (18 pts total): 3 points per panel (1 pt each for answers to (i), (ii), (iii), with justification required). Award full credit only if the physical or mathematical reasoning is stated, not just Yes/No.

Problem 2

[30 points]

Thick Spherical Shell: Gauss's Law to Potential

An insulating thick spherical shell has inner radius a, outer radius $b = 2a$, and uniform volume charge density $\rho > 0$. The cavity $r < a$ is empty.



(a) **[4 pts] Symmetry argument.**

Solution

The charge distribution is **spherically symmetric**: it is invariant under any rotation about the origin (since ρ is uniform throughout the shell and the geometry is perfectly spherical). By this symmetry, the electric field $\vec{E}$ cannot prefer any direction over another — it must be purely radial. Moreover, all points at the same r are equivalent by rotational symmetry, so E can depend only on r. Therefore:

$$\vec{E}(r) = E(r)\,\hat{r}$$

This symmetry also guarantees that $\vec{E}$ is constant in magnitude on any concentric sphere, which is precisely what we need to apply Gauss's law efficiently.

(b) **[10 pts] Electric field via Gauss's law.**

Solution

For each region, draw a concentric spherical Gaussian surface of radius r and apply:

$$\oint \vec{E} \cdot d\vec{A} = E(r) \cdot 4\pi r^2 = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

(i) $r < a$ (inside the cavity):
No charge is enclosed: $Q_{\text{enc}} = 0$.

$$\boxed{\vec{E} = \vec{0}}$$

(ii) $a \leq r \leq 2a$ (inside the shell material):
The enclosed charge is the charge in a spherical shell from radius a to r:

$$Q_{\text{enc}} = \rho \cdot \frac{4\pi}{3}(r^3 - a^3)$$

Applying Gauss's law:

$$E(r) \cdot 4\pi r^2 = \frac{\rho}{\varepsilon_0} \cdot \frac{4\pi}{3}(r^3 - a^3)$$

$$\boxed{\vec{E} = \frac{\rho(r^3 - a^3)}{3\varepsilon_0\, r^2}\,\hat{r}}$$

(iii) $r > 2a$ (outside the shell):
All charge is enclosed:

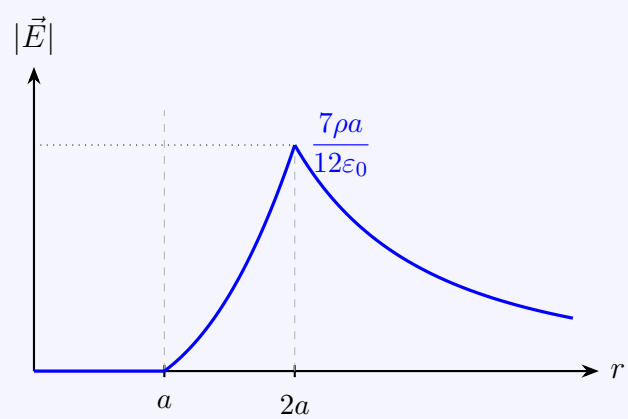
$$Q_{\text{tot}} = \rho \cdot \frac{4\pi}{3}\left[(2a)^3 - a^3\right] = \rho \cdot \frac{4\pi}{3}(7a^3) = \frac{28\pi\rho a^3}{3}$$

$$\boxed{\vec{E} = \frac{7\rho a^3}{3\varepsilon_0\, r^2}\,\hat{r}}$$

Values at the boundaries:

- At $r = a$: from region (ii), $E(a) = \dfrac{\rho(a^3 - a^3)}{3\varepsilon_0 a^2} = 0$ (*continuous* with region (i) ✓).
- At $r = 2a$: from region (ii), $E(2a) = \dfrac{\rho(8a^3 - a^3)}{3\varepsilon_0 \cdot 4a^2} = \dfrac{7\rho a}{12\varepsilon_0}$. From region (iii), $E(2a) = \dfrac{7\rho a^3}{3\varepsilon_0 \cdot 4a^2} = \dfrac{7\rho a}{12\varepsilon_0}$ (*continuous* ✓).

Sketch of $|E|$ vs. r:



(c) [**4 pts**] **Electric flux at $r = \frac{3}{2}a$.**

Solution

Method 1 — from the field:

$$\Phi_E = E\left(\tfrac{3a}{2}\right)\cdot 4\pi\left(\tfrac{3a}{2}\right)^2 = \frac{\rho\left[\left(\frac{3a}{2}\right)^3 - a^3\right]}{3\varepsilon_0\left(\frac{3a}{2}\right)^2}\cdot 4\pi\left(\tfrac{3a}{2}\right)^2 = \frac{4\pi\rho}{3\varepsilon_0}\left[\frac{27a^3}{8} - a^3\right] = \frac{4\pi\rho}{3\varepsilon_0}\cdot\frac{19a^3}{8} = \boxed{\frac{19\pi\rho a^3}{6\varepsilon_0}}$$

Method 2 — directly from Gauss's law:

$$Q_{\text{enc}} = \rho\cdot\frac{4\pi}{3}\left[\left(\frac{3a}{2}\right)^3 - a^3\right] = \frac{4\pi\rho}{3}\cdot\frac{19a^3}{8} = \frac{19\pi\rho a^3}{6}$$

$$\Phi_E = \frac{Q_{\text{enc}}}{\varepsilon_0} = \frac{19\pi\rho a^3}{6\varepsilon_0} \quad \checkmark$$

Both methods agree, as required by Gauss's law.

(d) **[8 pts] Electric potential** ($\varphi(\infty) = 0$).

Solution

We integrate $\varphi(r) = \int_r^\infty E(r')\,dr'$, working inward from ∞.

Region (i): $r > 2a$

$$\varphi(r) = \int_r^\infty \frac{7\rho a^3}{3\varepsilon_0\, r'^2}\,dr' = \frac{7\rho a^3}{3\varepsilon_0}\left[\frac{1}{r'}\right]_r^\infty = \boxed{\frac{7\rho a^3}{3\varepsilon_0\, r}}$$

Boundary value: $\varphi(2a) = \dfrac{7\rho a^2}{6\varepsilon_0}$.

Region (ii): $a \le r \le 2a$

$$\varphi(r) = \varphi(2a) + \int_r^{2a} \frac{\rho(r'^3 - a^3)}{3\varepsilon_0\, r'^2}\,dr' = \frac{7\rho a^2}{6\varepsilon_0} + \frac{\rho}{3\varepsilon_0}\int_r^{2a}\left(r' - \frac{a^3}{r'^2}\right)dr'$$

$$= \frac{7\rho a^2}{6\varepsilon_0} + \frac{\rho}{3\varepsilon_0}\left[\frac{r'^2}{2} + \frac{a^3}{r'}\right]_r^{2a} = \frac{7\rho a^2}{6\varepsilon_0} + \frac{\rho}{3\varepsilon_0}\left[\left(2a^2 + \frac{a^2}{2}\right) - \left(\frac{r^2}{2} + \frac{a^3}{r}\right)\right]$$

$$= \frac{7\rho a^2}{6\varepsilon_0} + \frac{\rho}{3\varepsilon_0}\left[\frac{5a^2}{2} - \frac{r^2}{2} - \frac{a^3}{r}\right] = \boxed{\frac{2\rho a^2}{\varepsilon_0} - \frac{\rho r^2}{6\varepsilon_0} - \frac{\rho a^3}{3\varepsilon_0\, r}}$$

Continuity check at $r = 2a$:

$$\varphi(2a) = \frac{2\rho a^2}{\varepsilon_0} - \frac{\rho(4a^2)}{6\varepsilon_0} - \frac{\rho a^3}{3\varepsilon_0(2a)} = \rho a^2\left(2 - \frac{2}{3} - \frac{1}{6}\right)\frac{1}{\varepsilon_0} = \frac{7\rho a^2}{6\varepsilon_0} \quad \checkmark$$

Region (iii): $r < a$
Since $E = 0$ in this region, φ is constant:

$$\varphi(r) = \varphi(a) = \frac{2\rho a^2}{\varepsilon_0} - \frac{\rho a^2}{6\varepsilon_0} - \frac{\rho a^2}{3\varepsilon_0} = \rho a^2\left(2 - \frac{1}{6} - \frac{2}{6}\right)\frac{1}{\varepsilon_0} = \boxed{\frac{3\rho a^2}{2\varepsilon_0}}$$

Continuity check at $r = a$: value matches from both sides ✓.

(e) **[4 pts] Limits and interpretation.**

Solution

(i) Far-field limit, $r \gg 2a$:
The total charge of the shell is:

$$Q_{\text{tot}} = \rho \cdot \frac{4\pi}{3}(7a^3) = \frac{28\pi\rho a^3}{3}$$

For $r \gg 2a$, we already found $\varphi(r) = \dfrac{7\rho a^3}{3\varepsilon_0 r}$. Rewriting:

$$\varphi(r) = \frac{7\rho a^3}{3\varepsilon_0 r} = \frac{Q_{\text{tot}}}{4\pi\varepsilon_0 r} \cdot \frac{28\pi\rho a^3/3 \cdot 4\pi\varepsilon_0}{3\varepsilon_0 \cdot 28\pi\rho a^3/3}$$

More directly: $\dfrac{Q_{\text{tot}}}{4\pi\varepsilon_0 r} = \dfrac{28\pi\rho a^3}{3 \cdot 4\pi\varepsilon_0 r} = \dfrac{7\rho a^3}{3\varepsilon_0 r}$ ✓

Physical meaning: Far from the shell, the detailed internal structure does not matter — the shell looks like a point charge Q_{tot} at the origin. This is a consequence of the shell theorem (and Gauss's law).

(ii) Value of $\varphi(0)$:

Since $E = 0$ inside the cavity ($r < a$), there is no work done moving a test charge from $r = a$ inward to $r = 0$. Therefore:

$$\varphi(0) = \varphi(a) = \frac{3\rho a^2}{2\varepsilon_0}$$

$\varphi(0)$ is **equal to** $\varphi(a)$. There is no potential difference across the empty cavity because the field is identically zero there.

Problem 3

[20 points]

Energy Stored in a Uniformly Charged Sphere

A solid sphere of radius R with uniform volume charge density ρ. The field is:

$$\vec{E}(r) = \begin{cases} \dfrac{\rho\, r}{3\varepsilon_0}\,\hat{r} & r < R, \\ \dfrac{\rho R^3}{3\varepsilon_0 r^2}\,\hat{r} & r > R. \end{cases}$$

(a) **[4 pts] Energy density.**

Solution

Using $u_E = \dfrac{\varepsilon_0}{2}E^2$:

Inside $(r < R)$:

$$u_E = \frac{\varepsilon_0}{2}\left(\frac{\rho r}{3\varepsilon_0}\right)^2 = \boxed{\frac{\rho^2 r^2}{18\varepsilon_0}}$$

Outside $(r > R)$:

$$u_E = \frac{\varepsilon_0}{2}\left(\frac{\rho R^3}{3\varepsilon_0 r^2}\right)^2 = \boxed{\frac{\rho^2 R^6}{18\varepsilon_0\, r^4}}$$

(b) **[12 pts] Total energy.**

Solution

$$U = \int_0^\infty u_E\, 4\pi r^2\, dr = U_{\text{in}} + U_{\text{out}}$$

Interior contribution $(0 \le r \le R)$:

$$U_{\text{in}} = \int_0^R \frac{\rho^2 r^2}{18\varepsilon_0} \cdot 4\pi r^2\, dr = \frac{4\pi\rho^2}{18\varepsilon_0}\int_0^R r^4\, dr = \frac{4\pi\rho^2}{18\varepsilon_0}\cdot\frac{R^5}{5} = \frac{2\pi\rho^2 R^5}{45\varepsilon_0}$$

Exterior contribution $(r \ge R)$:

$$U_{\text{out}} = \int_R^\infty \frac{\rho^2 R^6}{18\varepsilon_0\, r^4}\cdot 4\pi r^2\, dr = \frac{4\pi\rho^2 R^6}{18\varepsilon_0}\int_R^\infty r^{-2}\, dr = \frac{4\pi\rho^2 R^6}{18\varepsilon_0}\cdot\frac{1}{R} = \frac{4\pi\rho^2 R^5}{18\varepsilon_0} = \frac{2\pi\rho^2 R^5}{9\varepsilon_0}$$

Total:

$$U = \frac{2\pi\rho^2 R^5}{45\varepsilon_0} + \frac{2\pi\rho^2 R^5}{9\varepsilon_0} = \frac{2\pi\rho^2 R^5}{\varepsilon_0}\left(\frac{1}{45} + \frac{5}{45}\right) = \frac{2\pi\rho^2 R^5}{\varepsilon_0}\cdot\frac{6}{45} = \frac{4\pi\rho^2 R^5}{15\varepsilon_0}$$

Now substitute $Q = \frac{4}{3}\pi R^3 \rho$, so $\rho = \dfrac{3Q}{4\pi R^3}$ and $\rho^2 = \dfrac{9Q^2}{16\pi^2 R^6}$:

$$U = \frac{4\pi}{15\varepsilon_0} \cdot \frac{9Q^2}{16\pi^2 R^6} \cdot R^5 = \frac{36\pi Q^2}{240\pi^2 \varepsilon_0 R} = \boxed{\frac{3Q^2}{20\pi\varepsilon_0 R}}$$

(c) [**4 pts**] **Sanity checks.**

Solution

(i) Units of $Q^2/(\varepsilon_0 R)$:

$$\left[\frac{Q^2}{\varepsilon_0 R}\right] = \frac{\mathrm{C}^2}{\left(\frac{\mathrm{C}^2}{\mathrm{N}\cdot\mathrm{m}^2}\right)\cdot \mathrm{m}} = \frac{\mathrm{C}^2 \cdot \mathrm{N} \cdot \mathrm{m}^2}{\mathrm{C}^2 \cdot \mathrm{m}} = \mathrm{N} \cdot \mathrm{m} = \mathrm{J} \quad \checkmark$$

(ii) As $R \to 0$ with Q fixed:

$$U = \frac{3Q^2}{20\pi\varepsilon_0 R} \longrightarrow \infty$$

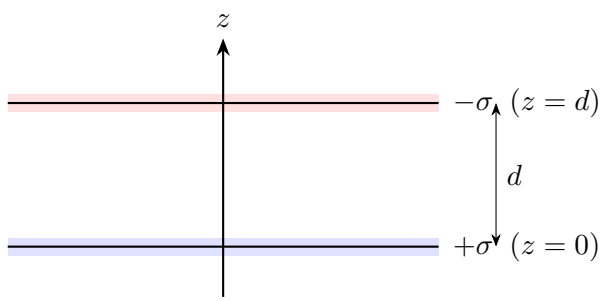
The energy diverges. This means that in classical physics, it would take *infinite* energy to assemble a point charge — a classical point charge has infinite self-energy. This is a fundamental difficulty of classical electrostatics, unresolved without quantum mechanics.

Problem 4

[17 points]

Two Infinite Parallel Sheets

Bottom sheet at $z = 0$: surface charge density $+\sigma$. Top sheet at $z = d$: surface charge density $-\sigma$.



(a) **[5 pts] Electric field by superposition.**

Solution

Each infinite sheet produces a uniform field of magnitude $\sigma/(2\varepsilon_0)$ pointing *away* from the sheet (for $+\sigma$) or *toward* it (for $-\sigma$):

Region	**From $+\sigma$ at $z = 0$**	**From $-\sigma$ at $z = d$**	**Total $\vec{E}$**
$z < 0$	$-\dfrac{\sigma}{2\varepsilon_0}\hat{z}$	$+\dfrac{\sigma}{2\varepsilon_0}\hat{z}$	$\vec{0}$
$0 < z < d$	$+\dfrac{\sigma}{2\varepsilon_0}\hat{z}$	$+\dfrac{\sigma}{2\varepsilon_0}\hat{z}$	$\dfrac{\sigma}{\varepsilon_0}\hat{z}$
$z > d$	$+\dfrac{\sigma}{2\varepsilon_0}\hat{z}$	$-\dfrac{\sigma}{2\varepsilon_0}\hat{z}$	$\vec{0}$

$$\boxed{\vec{E} = \begin{cases} \vec{0} & z < 0 \\ \dfrac{\sigma}{\varepsilon_0}\hat{z} & 0 < z < d \\ \vec{0} & z > d \end{cases}}$$

(b) **[4 pts] Electric potential** ($\varphi(0) = 0$).

Solution

Integrate $\varphi(z) = \varphi(0) - \displaystyle\int_0^z \vec{E} \cdot d\vec{z}'$ in each region:

$z < 0$**:** $E = 0$, so $\varphi = \varphi(0) = 0$. $\quad \varphi(z) = 0$.

$0 < z < d$**:**

$$\varphi(z) = 0 - \int_0^z \frac{\sigma}{\varepsilon_0}\, dz' = -\frac{\sigma z}{\varepsilon_0}$$

$z > d$**:** $E = 0$, so φ is constant at its value from $z = d$:

$$\varphi(z) = \varphi(d) = -\frac{\sigma d}{\varepsilon_0}$$

$$\boxed{\varphi(z) = \begin{cases} 0 & z < 0 \\ -\dfrac{\sigma z}{\varepsilon_0} & 0 \le z \le d \\ -\dfrac{\sigma d}{\varepsilon_0} & z > d \end{cases}}$$

Continuity: At $z = 0$: $0 = 0$ ✓. At $z = d$: $-\sigma d/\varepsilon_0 = -\sigma d/\varepsilon_0$ ✓.

(c) [**4 pts**] **Field energy per unit area.**

Solution

Between the plates, $E = \sigma/\varepsilon_0$ (constant), so:

$$\frac{U}{A} = \int_0^d \frac{\varepsilon_0}{2} E^2\, dz = \frac{\varepsilon_0}{2}\left(\frac{\sigma}{\varepsilon_0}\right)^2 d = \boxed{\frac{\sigma^2 d}{2\varepsilon_0}}$$

(d) [**4 pts**] **Physical interpretation.**

Solution

(i) Outside both sheets ($z < 0$ and $z > d$), $\vec{E} = \vec{0}$.
By superposition, the $+\sigma$ sheet produces $+\sigma/(2\varepsilon_0)$ outward on both sides, and the $-\sigma$ sheet produces $\sigma/(2\varepsilon_0)$ *inward* on both sides. Outside the capacitor, these two contributions point in *opposite directions* along $\hat{z}$ and exactly cancel. This is why a parallel-plate capacitor confines its field entirely between the plates.

(ii) As $d \to 0$ with σ fixed:

$$\frac{U}{A} = \frac{\sigma^2 d}{2\varepsilon_0} \longrightarrow 0$$

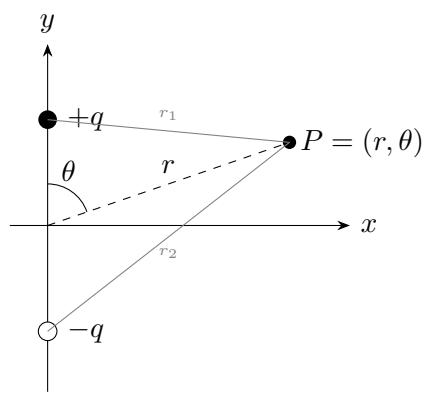
This is physically reasonable: as $d \to 0$, the $+\sigma$ and $-\sigma$ sheets overlap and cancel each other, producing no net charge and no electric field anywhere. With no field, there is no stored energy. The energy vanishes because the capacitance $C \propto A/d \to \infty$ while the stored charge goes to zero (the two surfaces neutralize).

Problem 5

[15 points]

The Electric Dipole

Charges $+q$ at $y = +\ell/2$ and $-q$ at $y = -\ell/2$; dipole moment $p = q\ell$. Field point P is at (r, θ) where θ is measured from the $+y$ axis.



(a) **[3 pts] Exact potential.**

Solution

By superposition, the potential at P is:

$$\boxed{\varphi_P = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_1} - \frac{1}{r_2}\right)}$$

where r_1 is the distance from $+q$ to P and r_2 is the distance from $-q$ to P.

(b) **[5 pts] Far-field approximation.**

Solution

For $r \gg \ell$, the given approximations give:

$$\frac{1}{r_1} \approx \frac{1}{r}\left(1 + \frac{\ell\cos\theta}{2r}\right), \qquad \frac{1}{r_2} \approx \frac{1}{r}\left(1 - \frac{\ell\cos\theta}{2r}\right)$$

Subtract:

$$\frac{1}{r_1} - \frac{1}{r_2} \approx \frac{1}{r}\left[\left(1 + \frac{\ell\cos\theta}{2r}\right) - \left(1 - \frac{\ell\cos\theta}{2r}\right)\right] = \frac{1}{r}\cdot\frac{\ell\cos\theta}{r} = \frac{\ell\cos\theta}{r^2}$$

Therefore:

$$\varphi(r,\theta) = \frac{q}{4\pi\varepsilon_0}\cdot\frac{\ell\cos\theta}{r^2} = \boxed{\frac{p\cos\theta}{4\pi\varepsilon_0\, r^2}}, \qquad p = q\ell$$

(c) [**4 pts**] **Far-field electric field.**

Solution

Apply $\vec{E} = -\nabla\varphi$ in polar coordinates:
Radial component:

$$E_r = -\frac{\partial\varphi}{\partial r} = -\frac{\partial}{\partial r}\left(\frac{p\cos\theta}{4\pi\varepsilon_0\, r^2}\right) = \frac{2p\cos\theta}{4\pi\varepsilon_0\, r^3}$$

Polar component:

$$E_\theta = -\frac{1}{r}\frac{\partial\varphi}{\partial\theta} = -\frac{1}{r}\cdot\frac{(-p\sin\theta)}{4\pi\varepsilon_0\, r^2} = \frac{p\sin\theta}{4\pi\varepsilon_0\, r^3}$$

Combining:

$$\boxed{\vec{E}(r,\theta) = \frac{p}{4\pi\varepsilon_0\, r^3}\Big(2\cos\theta\,\hat{r} + \sin\theta\,\hat{\theta}\Big)}$$

Therefore $A = 2$, $B = 1$.

(d) [**3 pts**] **Special directions.**

Solution

(i) $\theta = 0$ (on the $+y$ axis, far above $+q$):

$$\vec{E} = \frac{p}{4\pi\varepsilon_0\, r^3}(2\cdot 1\,\hat{r} + 0) = \frac{2p}{4\pi\varepsilon_0\, r^3}\,\hat{r}$$

At $\theta = 0$, $\hat{r} = \hat{y}$, so $\vec{E}$ points in the $+\hat{y}$ direction (away from $+q$, along the dipole axis). **Physically sensible:** far above the dipole, the $+q$ charge is closer and dominates; a positive test charge is repelled upward (in $+\hat{y}$).

(ii) $\theta = \pi/2$ (on the equatorial x-axis):

$$\vec{E} = \frac{p}{4\pi\varepsilon_0\, r^3}(0 + 1\cdot\sin\tfrac{\pi}{2}\,\hat{\theta}) = \frac{p}{4\pi\varepsilon_0\, r^3}\,\hat{\theta}$$

At $\theta = \pi/2$, the polar unit vector $\hat{\theta}$ points in the $-\hat{y}$ direction. So $\vec{E}$ points in the $-\hat{y}$ direction (antiparallel to the dipole moment). **Physically sensible:** on the equatorial plane, the $-q$ charge pulls a positive test charge downward (toward $-q$, i.e., in $-\hat{y}$). The two contributions from $+q$ and $-q$ reinforce each other in the $-\hat{y}$ direction there.

End of Answer Key. Total: 100 points.